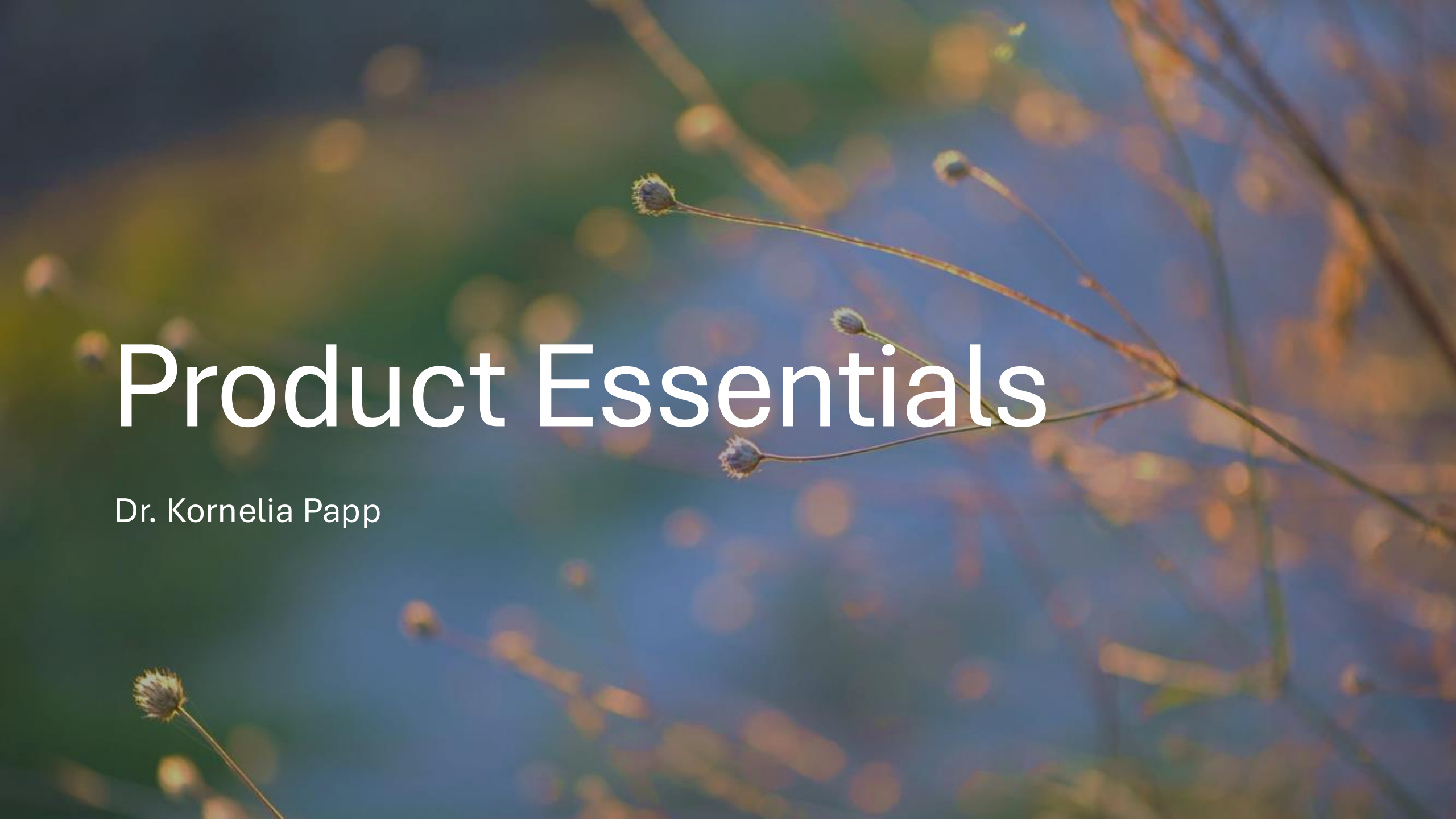
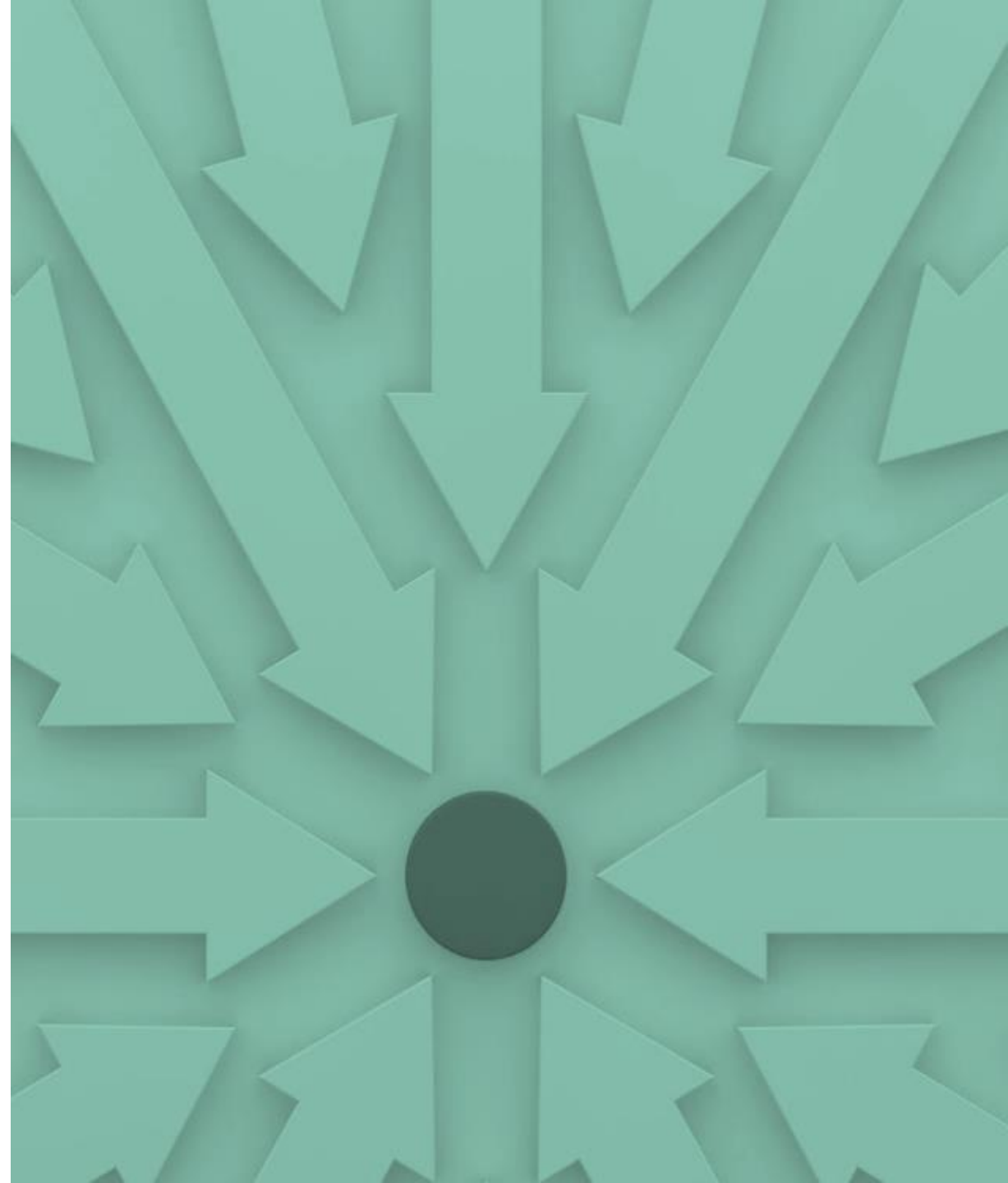




Product Essentials

Dr. Kornelia Papp

Building products
not (just) with AI
but with **purpose.**



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Session Goals

Understand how to move from an initial idea to a structured product opportunity by connecting **use case development**, **value assessment**, and **product development** into one coherent decision-making process.

What is being built?

Start with **use case development**: identifying the user, the core problem, the context, and the specific situation where AI or a product creates measurable value.

Should it be built?

Perform a **value assessment**: evaluating whether the use case is sufficiently important, feasible, and valuable to justify the investment of resources.

How is it built?

Move into **product development**: transforming the validated use case into a concrete concept by defining features, testing assumptions, and planning implementation.



Product development is rarely linear. Using AI to rapidly test and pivot is key.

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Use Case Development

Start with the
problem, not the
model.

User Need × AI Strength Framework

- Find good AI use cases before jumping to a tool
- A good AI use case is a real user problem × a task where AI has a structural advantage

“We have AI → where can we use it?” ❌

“We have a problem → does AI help?” ✅

- AI Strength: scale, speed, pattern
- Cost of failure: where does AI fail?
- Tolerance of error
- Human–AI complementarity
- **Don’t ask where to use AI. Ask where AI is structurally better than humans.**

Define User Need

What is the real user problem? Is an AI-based solution better than conventional software or process alternatives?

Question

Example

Who is the user?

e.g. student, doctor, customer-support agent

What is their painful task?

repetitive, uncertain, slow, cognitively heavy

What does success look like?

faster, cheaper, better, safer, more personalized

Why AI?

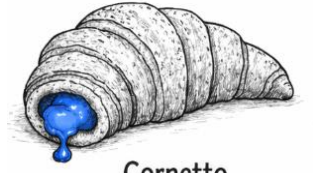
prediction, generation, classification, summarization, retrieval, automation

What should the human still control?

final decision, validation, exceptions



Croissant au Beurre
(French Classic)



Cornetto
(Italian Style)



Cronut



Cruffin



Flaffin



Gipfeli

Task Decomposition

A product is rarely a single item. Even a simple product is often a bundle of distinct components.

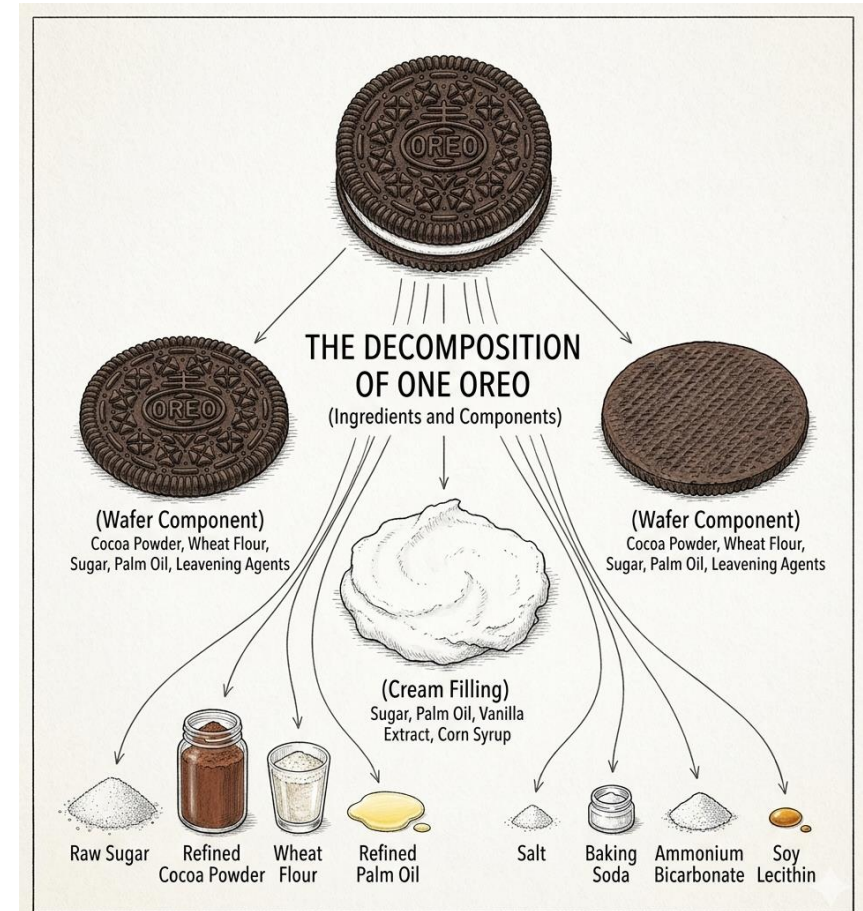
Forced decomposition: The purpose of decomposition is to find the *one* specific step where AI has a structural advantage.

Example: AI might be mediocre at "Writing a Roadmap," but it is world-class at "Extracting key points from 50 pages of notes."

Decomposition: Identify every discrete step required to produce the "Product."

Evaluation: Determine which sub-tasks are repetitive, high-volume, or data-heavy.

Integration: Apply AI only to the specific steps where it offers a structural advantage.



AI Capability Mapping

Indicators of a Strong AI Use Case

Target tasks where human constraints create a bottleneck:

- ✓ **High Frequency:** Tasks performed repeatedly where automation yields massive time savings.
- ✓ **Cognitive Drudgery:** Mentally taxing work that exhausts human focus but is trivial for a machine.
- ✓ **Probabilistic Logic:** Scenarios involving nuance and ambiguity that cannot be solved with rigid, "if-then" rules.
- ✓ **Massive Scale:** Data inputs or volumes that exceed human processing capacity.
- ✓ **Velocity Demands:** Situations requiring near-instant turnaround under significant time pressure.

Example: Report generation task

Task Step (Ingredients)	AI Fit / Structural Advantage
Reading large text	✓ Summarization
Extracting key points	✓ Pattern Recognition
Structuring output	✓ Generation
Verifying correctness	✗ Human-in-the-loop required

Human Role

Red Flags for a Weak AI Use Case

Avoid using AI when it adds unnecessary complexity or risk:

-  **Deterministic Tasks:** If a problem has a 100% predictable, rule-based solution, standard software is more efficient and reliable.
-  **Ungoverned High Stakes:** Safety-critical or high-liability decisions should never be fully automated without human oversight.
-  **Low Friction:** AI implemented for "novelty" that does not solve a documented user pain point.
-  **Vague ROI:** Projects lacking a clear, measurable benefit to the product's value or efficiency.

Example: Medical Diagnostic		
Phase	By	Explanation
Detection	AI	Handles massive input size and repetition that would cause human fatigue.
Prioritization	AI	Navigates ambiguity to surface only the most relevant or high-risk data points.
Verification	H	Ensures safety-critical decisions are governed by ethics and context, not just probability.
Decision	H	Manages scenarios where AI structural advantage fails or data is incomplete.

A Good AI Use Case

A good AI use case = A validated user pain point × A task where AI has a structural advantage.

1. **Identify the pain:** What is the specific bottleneck in the current "Product" (Code, Roadmap, Design, or Risk Model)?
2. **Verify the Advantage:** Is AI better here due to **scale**, **speed**, or **consistency**?
3. **Assess the Failure:** What happens if the AI is wrong? Is there a human-in-the-loop "Quality Gate"?
4. **Define the Complementarity:** Does using AI here free the human to focus on **strategy**, **empathy**, or **ethics**?

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Define Success

Quantifying Success

A use case without a metric is a weak use case. To move from a pilot to a product, success must be defined by measurable improvement.

Primary Metrics

Efficiency (Time Saved): Does the AI-assisted workflow reduce the end-to-end processing time for a specific task?

Accuracy (Error Reduction): Does the model effectively lower the rate of human error in high-volume processing?

Utility (Output Quality): Is the resulting product (e.g., a report or a diagnostic) more comprehensive or personalized than the manual alternative?



If success cannot be quantified, the use case is insufficient for development. Measurable benefits confirm structural advantage.

Value × Feasibility × Risk Prioritization Matrix

Which AI use case should you build first?

Framework	Best for	How to calculate value
1. ROI / NPV business case	Financial value	Estimate revenue gain, cost saving, risk reduction, then subtract build/run/change costs. Use NPV if benefits arrive over years.
2. RICE	Product / AI feature prioritization	Reach × Impact × Confidence / Effort. Very teachable and fast.
3. WSJF	Agile / portfolio sequencing	Cost of Delay / Job Size. Prioritize high urgency, low effort use cases.
4. Desirability–Viability–Feasibility	Innovation screening	Score: user need, business value, technical feasibility. Good first filter before detailed ROI.
5. Real Options	Uncertain tech / AI / R&D use cases	Value small experiments that create future choices: learn, scale, stop, pivot. Useful when ROI is unknown early.

Pick boringly useful before impressively futuristic.

Score & Classify

Score every idea from 1–5

Dimension	Ask
Value	Would this save time, money, improve quality, or unlock something new?
Feasibility	Do we have the data, tools, skills, and integration path?
Risk	Could errors, privacy issues, bias, or over-automation cause harm?
Time to impact	Can we test it quickly?

Classify ideas

Type	Decision
High value, high feasibility, low risk	Start here
High value, low feasibility	Research/prototype
Low value, high feasibility	Avoid “toy AI”
High risk	Needs governance or human-in-the-loop



Evaluate & Discuss

- Form groups of four
- Review all proposals (on next slide)
- Evaluate each proposal (template is available to help)
- Rank proposals in Canva board:
<https://canva.link/ig5yuxdr2l3lvuf>

Proposals

Project A: The Legacy Data Cleaner. An AI that automatically reformats old, messy actuarial spreadsheets into standardized databases.

Project B: The "Vibe" Bot. A chatbot for the internal HR portal that answers "how are you feeling?" with inspirational quotes.

Project C: Automated Claims Adjuster. An AI that automatically approves or denies medical claims under \$500 without any human review.

Project D: Customer Churn Predictor. A model that uses historical behaviour to predict which high-value clients are likely to leave in the next 30 days.

Template

Section	Answer
User	
Painful task	
Current workaround	
Task breakdown	
AI strengths applied	
Human in the loop	
Expected value	

Use Case Examples

The Legacy Data Cleaner		The "Vibe" Bot	
User	Actuary / Data Engineer	User	HR Manager / Internal Employee
Pain	Manual reformatting of inconsistent legacy spreadsheets	Pain	Low engagement on internal portals
Task	Standardizing data into clean, queryable databases	Task	Providing inspirational quotes based on user mood
AI strength	Pattern recognition + Data transformation	AI strength	Sentiment analysis + Text generation
Human role	Audit sample outputs for accuracy	Human role	Content moderation / Setting tone guidelines
Value	Days of manual cleaning reduced to minutes	Value	Minimal (Classic "Toy AI"—high feasibility, low business impact)

Use Case Examples

Automated Claims Adjuster		Customer Churn Predictor	
User	Claims Operations Team	User	Product Manager / Marketing Lead
Pain	Massive backlog of low-dollar medical claims	Pain	High-value clients leaving without warning
Task	Instant approval or denial of claims < \$500	Task	Identifying churn risk behavior patterns
AI strength	Classification + Anomaly detection	AI strength	Prediction + Large-scale data correlation
Human role	Critical Oversight: Handle appeals and edge cases	Human role	Designing and executing the intervention strategy
Value	Massive scale; Significant risk requires governance	Value	Revenue retention of high-value accounts

Use Case Ranking

Rank	Use Case	Classification	Primary Metric (Why)	Strategic Justification
1	A: Legacy Data Cleaner	Start Here	Efficiency (Time Saved)	High feasibility and high value; cleaning data is the "foundational" ingredient required before more complex AI can be built.
2	D: Churn Predictor	Research / Prototype	Utility (Output Quality)	High strategic value for revenue but relies on the clean data from Rank 1 to provide accurate predictions.
3	C: Automated Adjuster	High Risk	Accuracy (Error Reduction)	High value but requires a robust Human-in-the-Loop governance structure to manage safety-critical stakes.
4	B: HR "Vibe" Bot	Avoid "Toy AI"	No Measurable Benefit	While technically easy (high feasibility), it lacks a real user pain point and offers no clear ROI.

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AI-driven Product Development

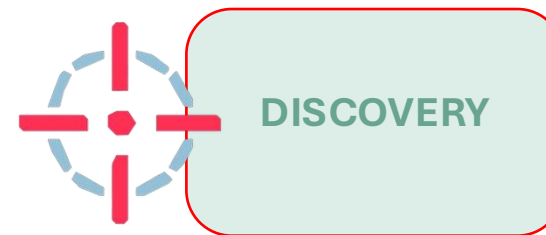
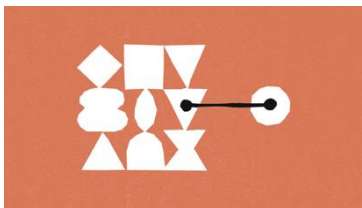
DISCOVERY

PROTOTYPE

ITERATE

EVALUATE

AI changes the process because outputs are probabilistic and quality depends on context, workflow, and trust.



Aspect		Traditional product development	AI-driven product development
 Process flow		Discovery → development → launch.	Discovery → prototype → evaluate → learn → iterate.
 Requirements		Requirements can be specified before build.	Behavior must be experienced before it is trusted.
 What drives design		Known happy path drives the design.	Unknown edge cases drive the learning agenda.
 Role of launch		Launch confirms the roadmap.	Launch starts the real feedback cycle.
 Quality focus		Quality mostly means bugs and UX friction.	Quality means correctness, usefulness, safety, trust.

Discovery

How should teams discover AI use cases?

Traditional product development

Interview users and validate needs.

Study pain points and unmet needs.

Ask whether users want the feature.

Prototype after the opportunity is clear.

Domain knowledge helps prioritization.

AI-driven product development

Do that, then add domain experts to define “good”.

Observe manual workflows where humans inspect, decide, summarize.

Ask whether AI can reliably help in the workflow.

Prototype while the opportunity is still being learned.

Domain knowledge is required for evaluation and prompts.

Workflow Role

Before building, decide the AI role in the workflow.

Traditional product development

Feature solves a step in the product.

User remains the main actor.

Tools sit inside fixed UI flows.

Success = task completed.

Mostly deterministic paths.

AI-driven product development

AI may **replace** a task: produce a summary or draft.

AI may **augment**: diagnose, suggest, explain, accelerate.

AI may be a **sidecar**: chat or assistant beside the workflow.

Success = user trusts the output enough to act.

Ambiguous roles need explicit boundaries and UX handoff.

Prototyping: From Figma to Interactive Prototyping

Traditional product development

PRD and mockups align the team.

Users can judge screens and flows.

Prototypes can use fake data.

Engineering starts after requirements.

Polish helps feedback.

AI-driven product development

Interactive AI prototypes reveal real behavior.

Users must judge outputs, timing, and failure modes.

Move quickly toward real context and real examples.

Engineers experiment early with prompts, context, and models.

Ugly but working prototypes are often better early.

Static mockups are not enough anymore. Show the model behavior early. Static slides hide the hard part.

Context is King

What does the model need to “see”?

Traditional product development

Context is mostly product state and user input.

Designers ask what screen to show.

More information rarely breaks simple features.

Success is predictable output.

Data access is implementation detail.

AI-driven product development

Context is the product: files, history, logs, permissions, intent.

Teams ask what context to retrieve and when.

Too much context can dilute relevance or exceed limits.

Success is the right information at the right moment.

Context strategy is core product strategy.

MVP

Why does MVP mean something different for AI?

Traditional product development

Thin slice can validate demand.

Users tolerate missing features.

MVP proves core workflow.

Launch earlier, learn from usage.

Failure is often visible as UX friction.

AI-driven product development

A weak AI output can destroy trust immediately.

Users may not tolerate wrong, random, or useless outputs.

AI MVP must be mature enough to evaluate fairly.

Launch carefully, then iterate continuously.

Failure can be silent: plausible but wrong answers.

Testing

How do you know the AI product is good?

Traditional product development

Use beta testing and A/B tests.

Measure adoption, conversion, retention.

QA checks expected paths.

Feedback after release is useful.

One release stabilizes the feature.

AI-driven product development

Add expert review, eval sets, and observed workflows.

Measure usefulness, correctness, trust, and saved effort.

Evaluation hunts long-tail edge cases.

Feedback loops are part of the product architecture.

Model/prompt/context changes keep the feature alive.

Useful before Impressive

Traditional product development

Ask: what user problem?

Map a feature in the flow.

Define required inputs.

Mock up the UI.

Check bugs and usability.

Roadmap after launch.

AI-driven product development

Ask: what human task can AI improve?

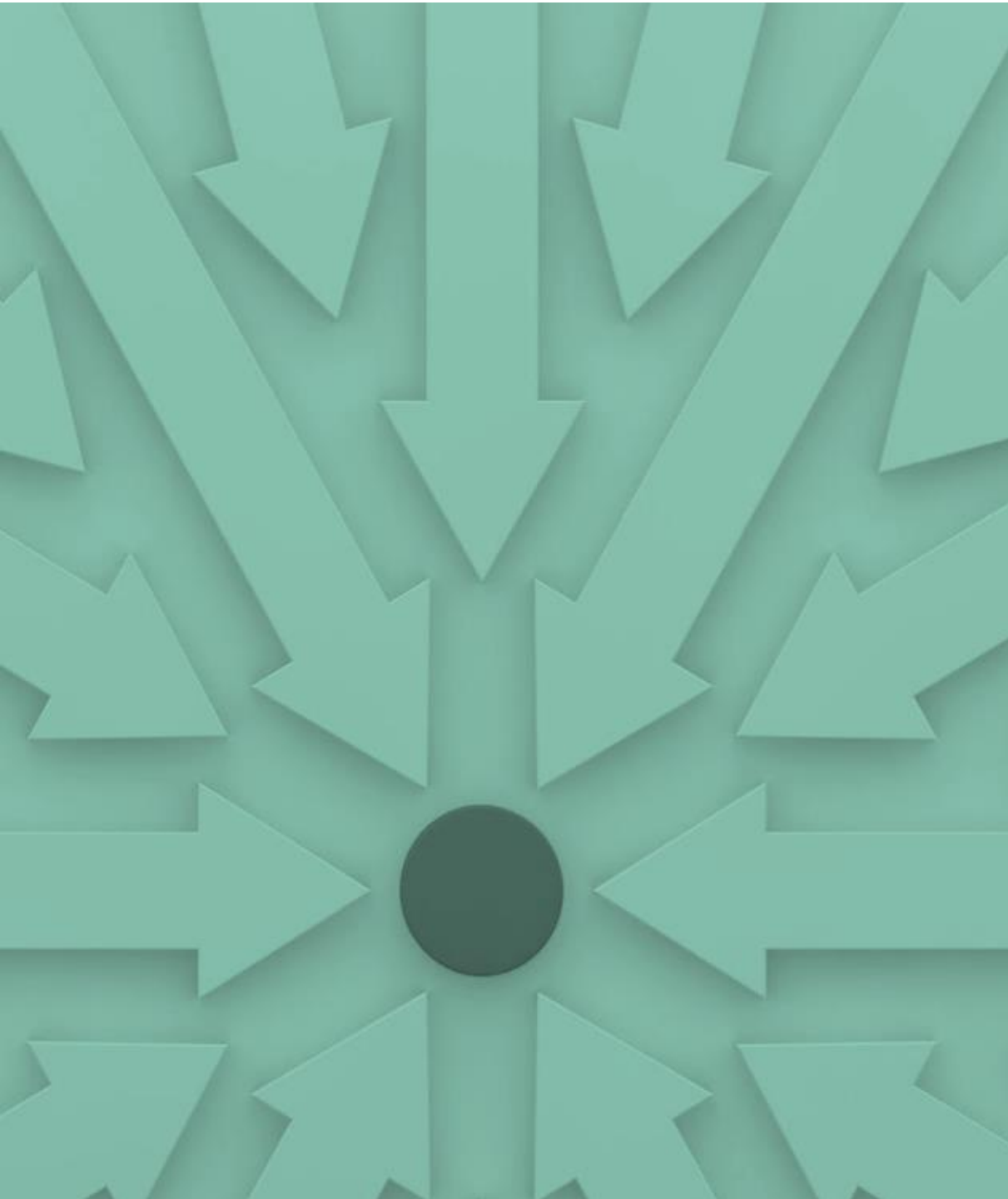
Choose AI role: replace, augment, or sidecar.

Define context strategy: what must the model know now?

Test real model behavior with real examples.

Check correctness, trust, and long-tail failures.

Feedback loop built in from day one.



Use AI to **shorten
the loop** between
building and
learning.

Let's Talk about the Hype

What is hype?



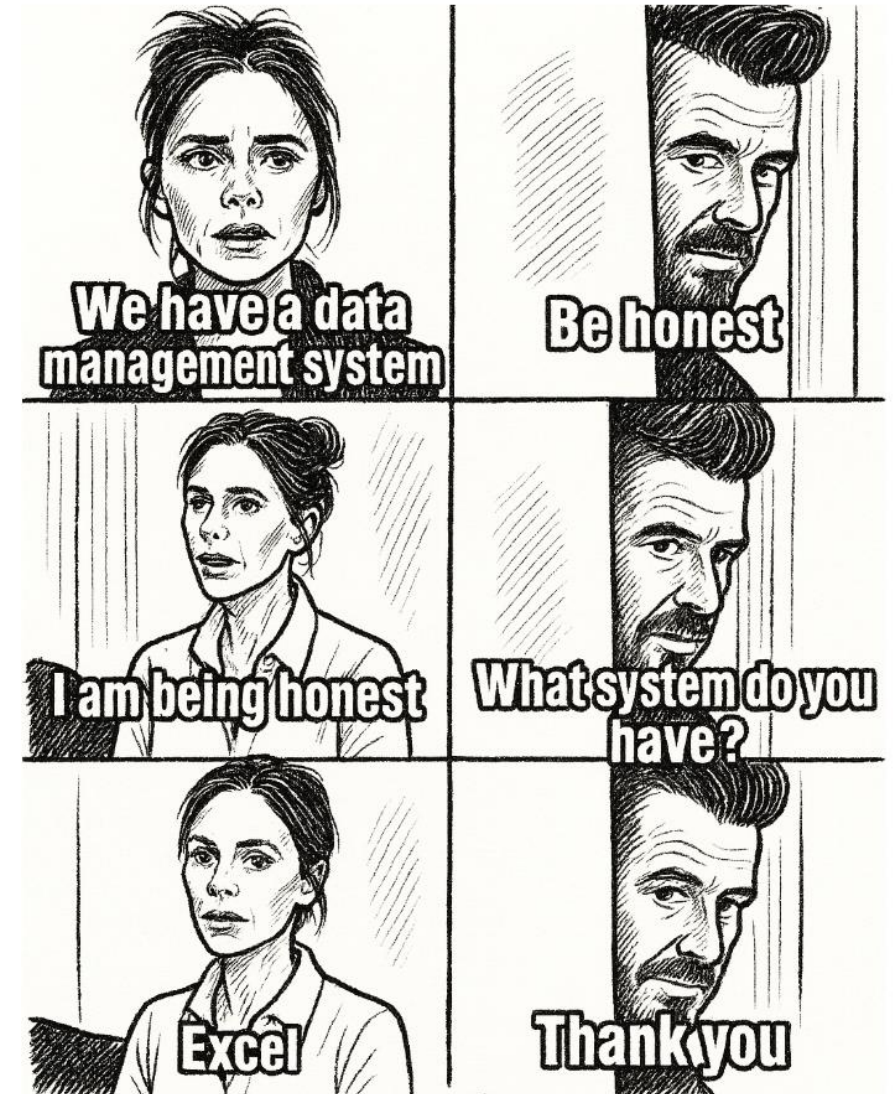
Urban Dictionary

<https://www.urbandictionary.com/define?term=hype>

hype

A fad. A clever marketing strategy which a product is advertized as the thing everyone must have, to the point where people begin to feel they need to consume ...

If you cannot manage hype, you will fail at your job

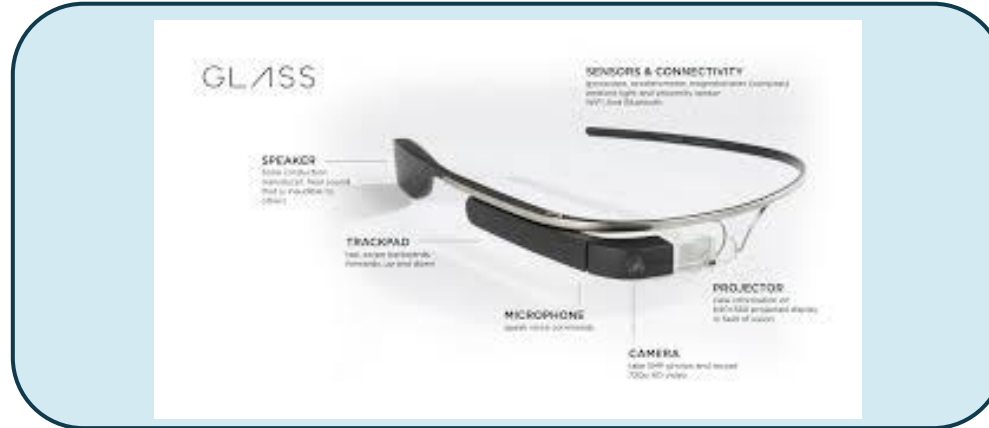


Hype

Apple AirPower



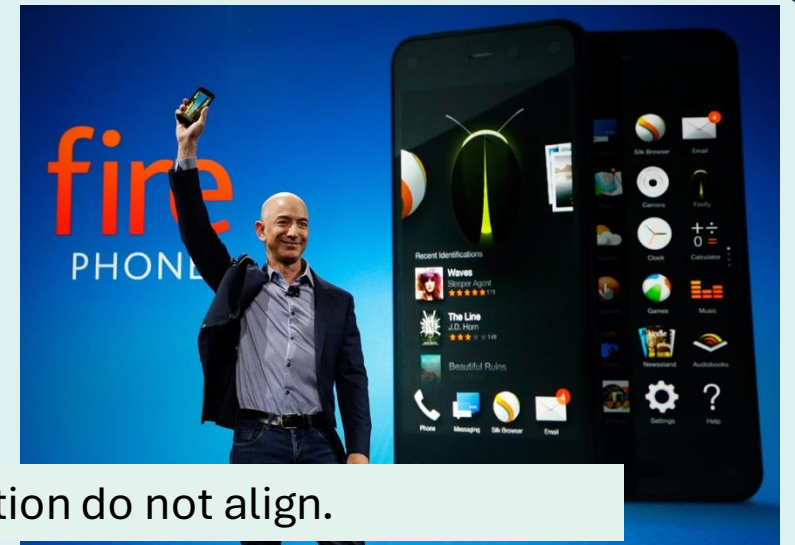
Google Glass consumer version



Apple Car / Project Titan



Amazon Fire Phone



Products fail when promise, timing, feasibility, and adoption do not align.

Why Products Fail

Reason for failure

Engineering was too hard

Scientific assumption failed

Regulation or political resistance

Missing ecosystem

Weak product-market fit

Social acceptance problem

Too expensive for the value offered

Internal company strategy conflict

Too early for the market/infrastructure

Physical-world complexity underestimated

Users did not trust long-term support

Example products

Apple AirPower; Google Project Ara; Magic Leap One

Google/Verily glucose-sensing smart contact lens

Facebook/Meta Libra/Diem; Apple Car

Amazon Fire Phone; Google Stadia; Microsoft Kin

Google Wave; Google Glass; Amazon Fire Phone;
Google Stadia

Google Glass; Facebook/Meta Libra/Diem

Google Glass; Microsoft Kin; Amazon Fire Phone

Microsoft Courier; Apple Car; Google Project Ara

Google Glass; Magic Leap One; Google Stadia

Apple Car; Google Project Ara; Magic Leap One

Google Stadia; Google Wave; Google Glass